MILITARY SPECIFICATION

TELESCOPE, OFFSET: XM121

1. SCOPE

1.1 This specification covers one type of 3-power, hand-held, night-viewing device which provides a 19-degree field of view.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army Munitions Command

F8565566	Pressure Tester-Two Station
F8578100	Telescope, XM121
IEL 8578100	List of Inspection Documents for Telescope, Offset, XM121

FSC 1240

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PACKAGING DATA SHEET

MIL-P-14232/P8578100 Packaging of Telescope, XM121

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The telescope shall be manufactured in accordance with Drawing F8578100 and drawings pertaining thereto.

3.2 General specification.- The contractor shall be responsible for adherence to, and compliance with, the requirements of MIL-F-13926 specified below:

(a) Order of precedence.
(b) Dimensions and tolerances.
(c) Inorganic protective surface finishes.
(d) Part identification and marking.
(e) Workmanship.
(f) Soldering electrical connections.

3.3 Environmental.

3.3.1 Storage temperatures.- The telescope shall show no evidence of physical failure such as cement separation, glass breakage, bonding failure, or casting failure due to storage at temperatures ranging from -65° to +155°F.

3.3.2 Extreme operating temperatures.- The telescope shall meet the applicable requirements of 3.13 while exposed to and thermally stabilized at temperatures of -40° and +150°F.

3.3.3 Standard temperature.- Unless otherwise specified herein, the telescope shall meet the following requirements while exposed to a temperature between 60° and 90°F.

3.3.4 Vibration.- The telescope shall show no evidence of physical failure due to being vibrated at 30 cycles per second and an amplitude of 0.06 (0.12 inch total excursion). Subsequent to meeting this requirement the telescope shall be capable of meeting the following requirements of this specification.

3.4 Sealing.- The telescope shall show no evidence of leakage when subjected to an internal pressure of 5 ± 0.10 pounds per square inch gage (psig) for a period of 5 minutes ± 15 seconds.

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3.5 Cleanliness.- The telescope shall meet the cleanliness requirements of MIL-O-13830 and there shall be no more than two particles of dirt visible through the eyepiece. The area of each particle shall not be in excess of the area equivalent to a size 10 dig. There shall be no foreign matter within the telescope which interferes with observation of adversely affects image quality.

3.6 Image intensifier tube.- The image intensifier tube shall conform to the applicable purchase description (see 6.3).

3.7 Image steadiness.- There shall be no visible flicker or movement of the image. This requirement shall be met with a positive potential of 5.5 $\pm$ 1.0 vdc applied to the oscillator and subsequent to allowing a 15 second time lapse to permit stabilization of the tube.

3.8 Image tilt.- The image of a vertical target shall be vertical within 2 degrees of arc.

3.9 Resolution.- Using no more than 10^{-3} lamberts to illuminate a resolving test chart defined in MIL-O-13830, the resolving power of the telescope shall be at least "X" minutes. "X" shall be derived from the following:

$$X \text{ minutes} = \frac{50}{\ell\text{-p/mm}}$$

where ℓ-p/mm is the line pairs per millimeter at the photocathode surface of the image tube which can be resolved on the image screen. The ℓ-p/mm is specified in the applicable purchase description.

3.10 Excursion range.

3.10.1 Ranging adjustment.- Rotation of the range adjustment ring shall cause the objective to bring targets from 5 meters to infinity into focus.

3.10.2 Diopter adjustment.- Rotation of the diopter knob shall provide for diopter settings throughout a range from -4 diopters to +4 diopters.

3.11 Backlash (ranging adjustment).- Backlash in the ranging adjustment mechanism shall not exceed 1/8 inch as measured on the outer diameter of the range adjustment ring.

3.12 Diopter setting.- The diopter knob shall indicate zero diopter within 1/8 diopter.

3.13 Operability.

3.13.1 Electrical switch.- The electrical switch shall be manually operable at the standard temperature specified in 3.3.3 and the extreme operating temperatures specified in 3.3.2.

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3.13.2 Ranging ring.- The running torque required to rotate the ranging ring at standard temperature shall be not less than 22 oz-in. and not more than 80 oz-in. The running torque required at the extreme operating temperatures shall be not less than 22 oz-in. and not more than 11 lb-in.

3.13.3 Diopter knob.- The running torque required to rotate the diopter knob at standard temperature shall be not less than 14 oz-in. and not more than 22 oz-in. The running torque required at the extreme operating temperatures shall be not less than 14 oz-in. and not more than 42 oz-in.

3.14 Interior atmosphere.- The interior atmosphere of the accepted telescope shall contain nitrogen having a dew point at least as low as -25°F.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection of components and sub-assemblies not covered by specifications or Supplementary Quality Assurance Provisions (SQAP) shall be in accordance with 4.2.2. Definitions of inspection terms not otherwise defined herein shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Inspection lot.- Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926 and the Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents, IEL 8578100. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with MIL-STD-105, and the inspection level and sampling plan specified in Table 1 titled "Classification of Defects". Examinations and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

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TABLE I - CLASSIFICATION OF DEFECTS

Use Inspection Level II of Table I with
Sampling Plan Table II-A of MIL-STD-105

		REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE			
MAJOR: AQL 0.65 percent defective			
101.	Vibration	3.3.4	4.5.4
102.	Cleanliness	3.5	4.5.7
103.	Image steadiness	3.7	4.5.8
104.	Image tilt	3.8	4.5.9
105.	Resolution	3.9	4.5.10
106.	Range adjustment	3.10.1	4.5.11
107.	Diopter adjustment	3.10.2	4.5.12
108.	Backlash (ranging adjustment)	3.11	4.5.13
109.	Diopter setting	3.12	4.5.14
110.	Electrical switch operability (60° to 90°F)	3.13.1	4.5.15
111.	Ranging ring (torque) (60° to 90°F)	3.13.2	4.5.16
112.	Diopter knob (torque) (60° to 90°F)	3.13.3	4.5.17
MINOR: NONE			

NOTE: The test in Table I shall be conducted at temperatures between 60° and 90°F.

4.2.3 Disposition of non-conforming product.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3 Control test.

4.3.1 General.- One telescope shall be selected at random as a control sample from each 100 produced, or from each months production, whichever occurs first. The control sample shall have successfully met all other acceptance tests specified herein, and shall then meet the following requirements and test:

CONTROL TEST		REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Visual, per drawing
302.	General specifications	3.2	Visual, MIL-F-13926

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4.3.2. Environmental.- Three telescopes shall be selected at random as a control sample from each 50 produced, or from each months production, whichever occurs first. The control samples shall conform to the following requirements and tests, and shall then meet all the requirements and tests tabulated in Table I:

CONTROL TEST	REQUIREMENT	TEST PROCEDURES
303. Storage temperatures	3.3.1	4.5.2
304. Electrical switch operability (-40° & + 150°F)	3.3.2 & 3.13.1	4.5.3 & 4.5.15
305. Range ring (torque) (-40° & +150°F)	3.3.2 & 3.13.2	4.5.3 & 4.5.16
306. Diopter knob (torque) (-40° & +150°F)	3.3.2 & 3.13.3	4.5.3 & 4.5.17
307. Sealing	3.4	4.5.5
308. Interior atmosphere	3.15	4.5.6

4.3.3 Control test failure.- Should any one item of a control test sampling fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall maintain all required measuring and testing equipment in accordance with the applicable requirements specified in MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.4.2 Contractor furnished inspection equipment.

4.4.2.1 Government design.- All inspection equipment required by drawings forming a part of the contract and not provided by the Government shall be supplied by the contractor in accordance with the requirements of 4.4 and the List of Inspection Documents IEL 8578100.

4.4.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" requirements of MIL-F-13926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

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4.4.2.3 Inspection equipment design requirements.

4.4.2.3.1 Sealing.- Special testing equipment to test sealing of the telescope shall conform to the internal pressure test requirements of Sheet 1 of Drawing F8565556, Pressure Tester - Two Station.

4.5 Test methods and procedures.

4.5.1 Image intensifier tube.- Inspection records shall be available to insure that the image intensifier tube has met the requirements of 3.6. In addition, the contractor shall provide evidence that a visual examination for defects or damage had been conducted prior to insertion of the tube into the telescope.

CAUTION: Do not direct the telescope toward the sun when the image intensifier tube is energized, because excessive light will damage this component.

4.5.2 Storage temperature.- Testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.3.1. The telescope, less battery, shall be exposed to each specified temperature extreme for a minimum of five hours. After the temperature exposures, the telescope shall be stabilized in an ambient temperature between +60° and 90°F and subjected to the audible, tactile and visual examination of 4.5.4. The telescope shall conform to the requirements of 3.3.1. Caution shall be exercised during this test to avoid subjecting the telescope to thermal shock.

4.5.3 Extreme operating temperature.- Testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.3.2. The telescope shall be exposed to each specified temperature extreme for a minimum of five hours. At the end of each five hour period and while at the temperature extreme, the electrical switch, ranging ring and diopter knob shall be subjected to the tests in 4.5.15 through 4.5.17 and shall conform to the operability requirements for each temperature extreme. Caution shall be exercised during this test to avoid subjecting the telescope to thermal shock.

4.5.4 Vibration.- Standard vibrating equipment capable of providing the amplitude and frequency specified in 3.3.4 shall be utilized for the vibration test. Unless otherwise specified, the test equipment shall be in conformance with the "Test Facilities" requirements of MIL-F-13926. The telescope shall be vibrated in a vertical plane for a period of 5 minutes $\pm$ 15 seconds. Subsequent to vibration the telescope shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine compliance with 3.3.4, prior to being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts. Minimum visual examination shall include the head assembly, body assembly and oscillator. A tactile examination should also be conducted

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to determine looseness or improper function of the above items and supporting or assembled members. An audible examination shall be made by positioning the telescope close to the ear and vigorously shaking it with both hands while listening for loose internal optical and mechanical parts (lenses, window, prisms, cells, retainers, adapters, tube, etc.). Any one of the possible defects defined above shall be cause to consider a telescope defective, unless separate defects-per-hundred type sampling plans are established and approved by the Government as part of the suppliers inspection system. Telescopes which have passed the above examinations shall be subjected to the tests of 4.5.7 through 4.5.17 inclusive to determine overall compliance with 3.3.4.

4.5.5 Sealing.- The internal pressurizing of the telescope, to test sealing, shall be performed with special testing equipment conforming to 4.4.2.3.1. Pressurize the telescope head and body assemblies, with dry nitrogen gas to the value specified in 3.4, in accordance with the procedure for internal pressure testing on Sheet 1 of Drawing F8565556. Retain the applied pressure in the telescope assemblies for a period of 5 minutes $\pm$ 15 seconds. During this time period, the ranging adjustment ring and the diopter knob of the telescope shall be rotated one full excursion clockwise and counterclockwise. At the end of the 5 minute period there shall be no evidence of leakage, as indicated on the pressure gauge of the testing equipment, to determine compliance with 3.4. The telescope shall then be subjected to the test in 4.5.6.

4.5.6 Interior atmosphere.- A dew-point tester shall be used to test the dew-point of the interior atmosphere of the telescope. Upon satisfactory completion of the sealing test, the internal pressure of the telescope head and body assemblies, shall be reduced to 1 psig. The assemblies shall then be flushed with dry nitrogen gas. The flow of gas into the assemblies shall be controlled by a regulator that will prevent the internal pressure in the assemblies from exceeding 1 psig during the flushing operation. The flushing shall continue until the nitrogen gas bled from the assemblies has a dew-point at least as low as minus 25°F, as measured with the dew-point tester. The assemblies shall then be sealed. The internal atmosphere of the head and body assemblies of the telescope shall conform to the requirements of 3.14.

4.5.7 Cleanliness.- Energize the image tube of the telescope with the voltage required in 3.7. The voltage shall be applied to the oscillator from a controlled power source that is capable of supplying and maintaining the required voltage. Visually inspect the optical system and image screen in the telescope for cleanliness, while the line of sight of the telescope is directed to a plain light backdrop that will cause uniform illumination of the image tube screen. Cleanliness of the telescope shall conform to the requirements of 3.5. A size 10 dig is equal to 0.1 millimeter.

4.5.8 Image steadiness.- Secure the telescope in a holding device. Direct the line of sight of the telescope to a grid pattern target, and position an auxiliary telescope at the eyepiece of the telescope. The auxiliary telescope shall contain a crossline reticle, and be capable of focusing on the screen of the image tube in the telescope. Energize the image tube of the telescope as in 4.5.7. Ob-

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serve through the auxiliary telescope and align the reticle pattern to the grid pattern image on the screen of the image tube. Immediately following the 15 second tube stabilization period, observe the grid pattern image for a time period of not less than one minute. During this one minute time period there shall be no visible flicker or movement of the image as specified in 3.7. Image movement shall be determined by means of the grid pattern image with reference to the telescope reticle pattern.

4.5.9 Image tilt.- Perform this test with the telescope positioned on a level surface. The telescope shall be secured, in the position of intended use, by means of appropriate "V" blocks that shall engage the cylindrical surfaces of the housing and body assembly. Appropriate non-abrasive cushioning material should be placed between the telescope and engagement surfaces of the "V" blocks to prevent damage to the telescope. The "V" blocks shall hold the telescope in a vise like manner. Direct the line of sight of the telescope to a target appropriately calibrated for measuring image tilt. Align an auxiliary telescope to the exit pupil (eyepiece) of the telescope by means of a suitable holding device. The vertical reticle line of the auxiliary telescope shall be plumb, and maintained plumb throughout this test. Energize the image tube of the telescope. Observe through the auxiliary telescope, and position the auxiliary telescope so that one extremity of its vertical reticle line is coincident to the target plumb line image. The image of the target plumb line shall be parallel to the vertical reticle line of the auxiliary telescope within the tolerance specified in 3.8, as measured on the target.

4.5.10 Resolution.- This test shall be performed utilizing a resolving power chart, on a wall target representing the angular subtense for the minutes of arc specified in 3.9. The resolving power chart shall be placed at the calculated distance from the front surface of the head prism of the telescope. Construction and location of the resolving power chart shall be as specified in the "Resolution test" of MIL-O-13830, and shall contain four line sets as shown in Figure I therein. The target shall be illuminated with white light. The intensity of the white light shall be reduced by means of neutral density filters, so that the light incident to the target does not exceed 1×10^{-3} lamberts. Position the telescope as in 4.5.9, and adjust the diopter knob for best focus of the phosphor grain on the screen of the image tube. The target image shall be viewed through the telescope eyepiece with the unaided eye. Obtain best focus of the numeral in the center of the resolving power chart by means of the ranging adjustment ring of the telescope. There shall be no further adjustment of the ranging ring or diopter knob for this test. At this point of focus of the telescope, all four meridians of the resolving power chart shall have the detectable line structure and line count to conform to the requirements of 3.9.

4.5.11 Ranging adjustment.- The test for short range focus requirements of the telescope shall be performed with an appropriate wall target positioned at 5 meters $\pm$ 0.3 meter. The telescope shall conform to the requirements of 3.10.1 for the short range. The test for infinity focus (long range) requirements of the telescope shall be performed with an appropriate target in a collimator projector, that is adjusted for infinity focus. The target shall be projected by means of white light. The intensity of the white light shall be reduced by means of neutral density

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filters, so that the light incident to the head prism does not exceed 1 X 10^1 lamberts. Align the optical axis of the telescope to the mechanical axis of the collimator projector, and exclude all ambient light incident to the head prism of the telescope. Adjust the range adjustment ring for best focus of the target. The telescope shall conform to the requirements of 3.10.1, for infinity focus.

4.5.12 Diopter adjustment.- Position the telescope diopter knob to zero diopter as indicated on the knob scale with reference to the index mark. Rotate the diopter knob to its clockwise stop, note diopter setting on the knob scale, then rotate the diopter knob to its counterclockwise stop, note diopter setting on the knob scale. The diopter knob shall conform to the requirements of 3.10.2, as indicated on the diopter knob scale.

4.5.13 Backlash (ranging adjustment).- Perform this test with a plumb line target positioned at the distance specified in 4.5.10. With backlash removed for direction of travel, obtain best focus of the plumb line target image. At this point of focus scribe a pencil mark on the ranging adjustment ring and the telescope housing. The marks shall be vertical and their adjacent extremities shall coincide. With backlash excluded, continue rotation of the ranging adjustment ring to its stop, then with backlash included, refocus on the plumb line target without overtravel. Any separation between the scribed pencil marks (lack of coincidence) at this focus position shall be measured with appropriate standard measuring equipment on the outer diameter of the ring. Any backlash in the ranging adjustment ring shall not exceed the tolerance specified in 3.11. Should overtravel occur during rotating of the ring, the test is nullified. When overtravel occurs the test for backlash shall be repeated.

4.5.14 Diopter setting.- Set the diopter knob of the telescope to indicate zero diopters. A calibrated dioptometer having a magnification of at least three power (3X) shall be utilized for measuring the accuracy of zero diopter setting of the diopter knob. Set the dioptometer diopter scale to zero, and adjust the dioptometer eyepiece for best focus of the dioptometer reticle. Position the dioptometer at the telescope eyepiece. While viewing through the dioptometer, obtain best focus of the phosphor grain on the screen of the image tube by adjustment of the telescope diopter knob. The phosphor grain on the screen of the image tube shall be in best focus within the tolerance specified in 3.12, as measured on the scale of the diopter knob.

4.5.15 Electrical switch operability.- The electrical switch of the telescope shall be subjected to a functional tactile examination. The electrical switch shall conform to the requirements specified in 3.13.1, for the temperatures indicated.

4.5.16 Ranging ring torque.- Special testing equipment that is capable of measuring the applied running torques required to rotate the ranging ring shall be utilized for this test. The running torque required to rotate the ranging ring shall be within the values specified in 3.13.2, for the temperatures indicated. The torque may be measured as the force applied tangent to the outer diameter of the ranging ring, provided there is no slippage between the testing equipment and ranging ring.

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4.5.17 Diopter knob torque.- Standard torque measuring equipment with a suitable adapter for engaging the diopter knob shall be utilized for this test. The running torque required to rotate the diopter knob shall be within the values specified in 3.13.3 for the temperatures indicated.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P8578100. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Telescope, XM121 is a part of the fire control system of the XM551 Vehicle and is intended for low light observation and surveillance.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of an applicable level of preservation, packaging, and packing.
(c) Applicable packaging data sheet number (see 5.1).
(d) Applicable stock number.

6.3 Image intensifier tube.- The detailed requirements for the image intensifier tube are specified in "Purchase Description for Image Intensifier Tube Assembly, (25 Millimeter) (U)" dated 3 April 1964. This purchase description is a classified document.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1240-A435Y

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